

# Existence and Uniqueness for the Sub-Riemannian Cubic Rhombus in Infinite Dimension (simply connected, smooth, mixed boundary conditions): Category-theoretic Proof

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## Abstract

In infinite dimension an infinite system of unit vectors with equal pairwise inner product  $c$  exists exactly for  $0 \leq c \leq 1$ , and its Gram matrix is a bounded operator on  $\ell^2$  only for  $c = 0$ . This manuscript states and proves exactly this proposition and nothing more. The title names a sub-Riemannian geometry, a simply connected topology, smooth regularity, mixed boundary conditions and a category-theoretic technique. None of these enters the argument (Section 4). The proposition belongs to class S09, which 254 manuscripts of this release share. This manuscript does not establish the result its title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 1 Introduction

The cubic rhombus  $R_n(c)$  is the parallelotope with unit edges and constant pairwise edge inner product  $c$ . Its Gram matrix is  $(1 - c)I + cJ$ , and every quantity below is computed from this one matrix. The classical background is the Gram determinant [1] and Hadamard's inequality [2]; see also [3]. This manuscript proves one proposition about this family. The title of the manuscript promises more than the proposition delivers. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 2 Setting

**Definition 1** (Cubic rhombus, finite sections, infinite systems). For an integer  $n \geq 2$  and  $c \in \mathbb{R}$  let  $G_n(c) = (1 - c)I_n + cJ_n$ , where  $J_n$  is the matrix of ones. A *cubic rhombus*  $R_n(c)$  is the parallelotope in  $\mathbb{R}^n$  spanned by unit vectors with pairwise inner product  $c$ . An *infinite system* is a sequence  $(v_i)_{i \geq 1}$  of unit vectors in a real Hilbert space with  $\langle v_i, v_j \rangle = c$  for  $i \neq j$ . Its finite sections are the systems  $(v_i)_{i \leq n}$ . The name “cubic rhombus” is used in this corpus for this family. It is not claimed to be standard terminology.

**Lemma 1** (Spectrum and volume).  $G_n(c)$  has eigenvalue  $1 + (n - 1)c$  once, with eigenvector  $\mathbf{1}$ , and eigenvalue  $1 - c$  with multiplicity  $n - 1$ . Hence  $\det G_n(c) = (1 - c)^{n-1}(1 + (n - 1)c)$ , positivity holds exactly for  $-\frac{1}{n-1} < c < 1$ , and then  $\text{vol}_n R_n(c) = \sqrt{\det G_n(c)}$ .

*Proof.*  $G\mathbf{1} = (1 + (n-1)c)\mathbf{1}$ , and  $Gx = (1-c)x$  for  $x \perp \mathbf{1}$ . If  $V$  has columns  $v_i$  then  $G = V^\top V$  and  $\text{vol}_n = \sqrt{\det G}$ .  $\square$

**Lemma 2** (Logarithmic derivative).  $F(c) = \ln \det G_n(c)$  satisfies  $F'(c) = -\frac{n(n-1)c}{(1-c)(1+(n-1)c)}$  on  $(-\frac{1}{n-1}, 1)$ .

*Proof.*  $F'(c) = -\frac{n-1}{1-c} + \frac{n-1}{1+(n-1)c}$ , and the common denominator gives the stated form.  $\square$

**Lemma 3** (Vertex Gram matrices). *At the vertex  $S$  of  $R_n(c)$  the Gram matrix of the unit edge directions is  $D_\varepsilon G_n(c) D_\varepsilon$  with  $\varepsilon_i = -1$  for  $i \in S$  and  $+1$  otherwise. If  $n \geq 3$  and all its off-diagonal entries equal a number  $s$ , then  $s = c$ .*

*Proof.* The off-diagonal entries are  $\varepsilon_i \varepsilon_j c$ . If  $c \neq 0$  put  $p = s/c$ . Then  $p^2 = \varepsilon_i \varepsilon_j \varepsilon_i \varepsilon_k = \varepsilon_j \varepsilon_k = p$  for distinct  $i, j, k$ , so  $p = 1$ .  $\square$

### 3 Result

**Theorem 1** (Existence in infinite dimension). *Let  $c \in \mathbb{R}$ . There is an infinite system of unit vectors with pairwise inner product  $c$  if and only if  $0 \leq c \leq 1$ . The infinite matrix  $G(c) = (1-c)I + cJ$  defines a bounded operator on  $\ell^2$  if and only if  $c = 0$ .*

*Proof.* Necessity: every finite section of order  $m \geq 2$  has a positive semidefinite Gram matrix, so  $1 + (m-1)c \geq 0$  by Lemma 1 for every  $m$ , which forces  $c \geq 0$ ; and  $c \leq 1$  by the Cauchy–Schwarz inequality. Sufficiency: put  $v_i = \sqrt{c}e_0 + \sqrt{1-c}e_i$  for an orthonormal sequence  $e_0, e_1, \dots$ . Then  $\|v_i\|^2 = c + (1-c) = 1$  and  $\langle v_i, v_j \rangle = c$  for  $i \neq j$ . Boundedness: the vector  $G(c)e_1$  has entry  $c$  in every coordinate except the first, so it lies in  $\ell^2$  only if  $c = 0$ ; and  $G(0) = I$ .  $\square$

### 4 Parameters carried in the title

| Parameter                     | Status    | Role in the argument                                                                                                      |
|-------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------|
| Dimension                     | Operative | The proposition is stated for the infinite dimensional case.                                                              |
| Contribution                  | Operative | The label corresponds to the proposition proved here.                                                                     |
| Geometry: sub-Riemannian      | Not used  | Every statement concerns the Euclidean metric on $\mathbb{R}^n$ .                                                         |
| Topology: simply connected    | Not used  | $R_n(c)$ is a convex polytope, hence contractible. No other topology is considered.                                       |
| Regularity: smooth            | Not used  | The object is a polytope. No function space on it is considered, and the label is carried as metadata.                    |
| Boundary conditions: mixed    | Not used  | No boundary value problem is posed. The word boundary in Theorem 1 refers to the facets of the polytope, where it occurs. |
| Technique: category-theoretic | Not used  | The proof given is by linear algebra (eigenvalues and a Cholesky factorization). The label is carried as metadata.        |

Of the seven parameters of the title, two enter the argument and five do not. The five that do not are retained so that the title is complete.

## 5 Relation to the rest of the release

The proposition of this manuscript is in class S09 (scope: infinite dimension). In this release, 254 manuscripts carry class S09. The principal member of this class in this release is manuscript CR-241-003. This manuscript restates that proposition under different title parameters. Counting manuscripts is therefore not counting results. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 6 Verification status

| Item                                               | Status        |
|----------------------------------------------------|---------------|
| Human review                                       | none          |
| Independent scrutiny                               | none          |
| Lean formalization                                 | not attempted |
| Exact computer algebra checks tagged to this class | 6 of 6 passed |
| Novelty of the proposition                         | not assessed  |

The computer algebra checks confirm closed forms against direct computation for  $n = 2, \dots, 8$  and confirm the symbolic identities in  $n$ . They do not address novelty or significance. Some results could have issues, and a correction would be recorded as a new version.

## 7 Statement of non-support

This manuscript proves a proposition about a determinant of a matrix of the form  $(1 - c)I + cJ$ . It does not prove, support, or provide evidence for the claim that the proposition is an independent result, a new result, or a result of the kind the title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## References

- [1] J. P. Gram, *Über die Entwicklung reeller Functionen in Reihen mittelst der Methode der kleinsten Quadrate*, J. Reine Angew. Math. 94 (1883), 41–73.
- [2] J. Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. 17 (1893), 240–246.
- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.